A usable pilot, but not yet a final PYQ-matched mock

Decision: revise before student release | Audit of the unchanged Test 1 snapshot

The paper meets its formal structure and has relevant, scientifically consistent diagrams. The main weaknesses are **style balance, difficulty calibration and a few weak distractors**. No definite wrong key was found under the stated assumptions in this review; that is not independent academic certification.

Parameter	Finding	Decision
Exam structure	60 questions; 15 per subject; 180 minutes; +4/-1/0	Pass for 2026 scheme
Syllabus mapping	All 60 map to retained families; combined subtopics need editorial sign-off	Broad fit; review detail
Chapter balance	Maths has 0 coordinate geometry versus 6/59 recent items	Revise
Question style	Chemistry is 9/15 numerical; PYQ coding is 12/60	Revise
Difficulty	Stored 9/39/12; this review suggests 18/32/10	Recalibrate content
Images	All six match their questions; four representation tasks, two setup sketches	Relevant; none clearly random
Keys and solutions	45 arithmetic probes pass; 60 explanations inspected	Pass checks; human review pending
Release readiness	No timed student trial or verified 2027 brochure	Not established

What to keep: strong linked reasoning in B13, C20, M40/M44 and P49/P54/P60; fixed IDs; separate solutions; exact graph and circuit drawing.

This is an AI-assisted self-audit by the assistant that authored the pilot, not an independent examiner’s certification. Difficulty and distractor quality are reasoned judgements; counts and checks are separately labelled.

What was compared, and against which standard?

Three different tests: official structure, project design rules and observed PYQ characteristics.

Benchmark	Evidence	How it is used
Official 2026 rules	Brochure sections 19/23 and official exam page	Counts, marking, time and calculator policy
OurMocks blueprint	Stored 2026 syllabus, fixed-order rule, 9/39/12 design mix	Internal authoring standard; not an official quota
Observed PYQ sample	539 items overall; recent 239; recent current-scope 231	Chapter, style and editorial difficulty comparisons
Current pilot	60 JSON items, six PNGs, 49-page review PDF	Item-level and image-level checks

How to interpret the comparisons

• Chapter reference counts are recent current-scope shares multiplied by 15. They can be fractional because they are a reference average, not a rule for one paper.

• Question-style comparisons use the same mutually exclusive dominant-task categories. The pilot was manually recoded, rather than treating its broad JSON format labels as equivalent.

• The audit proposes tiers from the reasoning actually left after the stem supplies its facts. These labels are not candidate pass rates and have not been validated by a student trial.

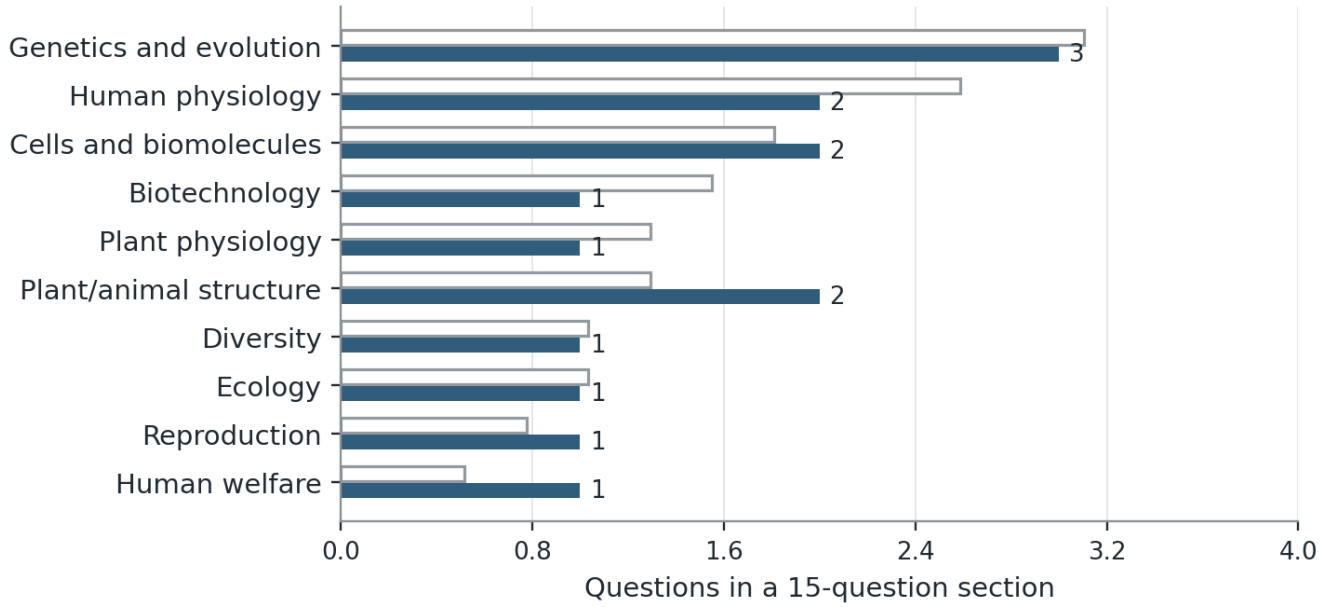
Snapshot identity

Canonical file: IISER/IAT TEST 1.json
Test ID: IAT27_TEST_01
SHA-256: f8a9fa3f78e36b1051789adc16893c87a8bae2837b8ab7e23ec6d19db1e43d4a

The audit does not rewrite, shuffle or renumber Test 1. Sources: repository ledger, taxonomy, plan, validator results, arithmetic probe results and render-check checksums. Full references and release actions appear on page 19.

Biology covers all ten broad units

Biology | Filled bars: Test 1. Open bars: recent current-scope share × 15.



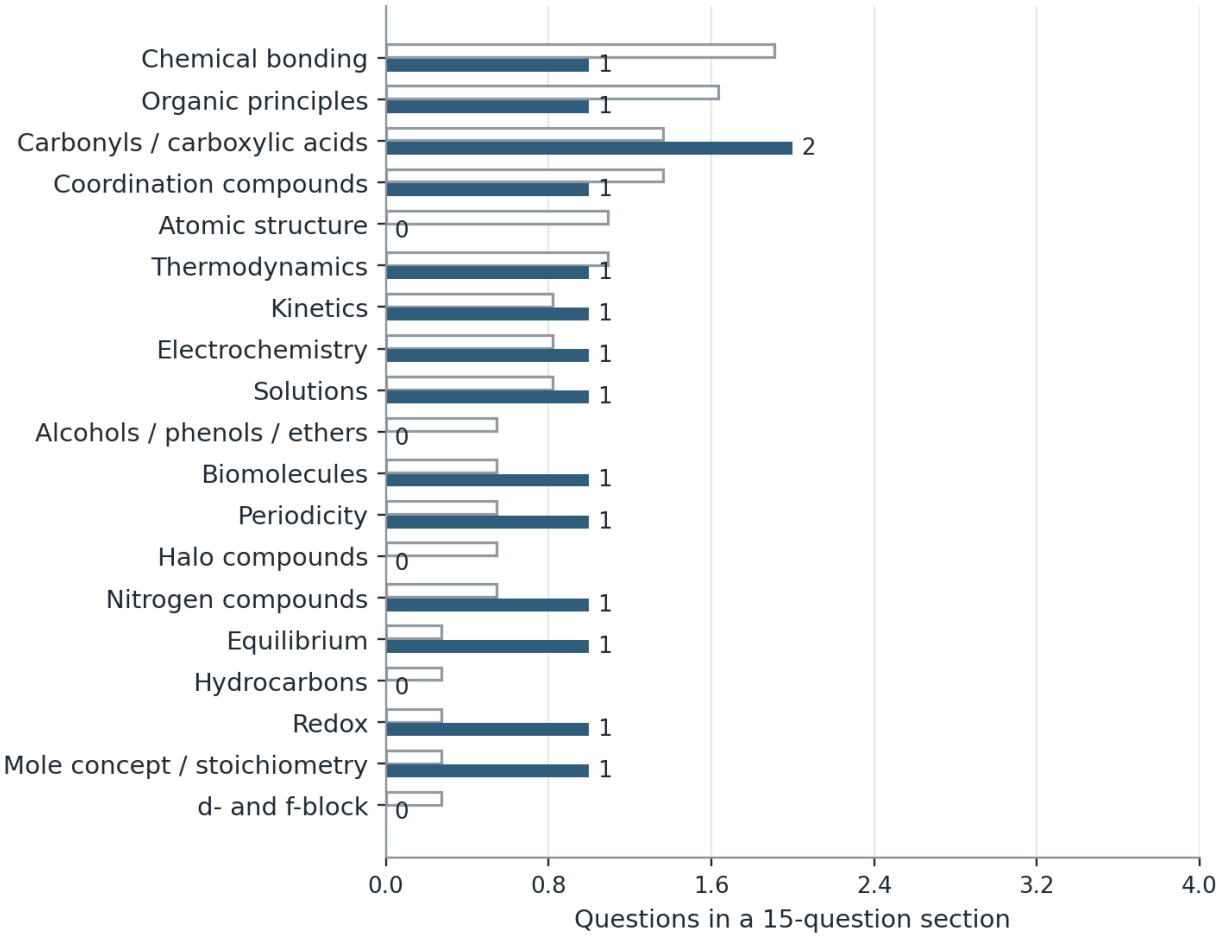
Denominator: 58 recent current-scope questions. Fractional references are not prescribed quotas. Each item counts once by its primary family.

All ten retained Biology families appear. Genetics has 3 questions; cell, human physiology and structure have 2 each; other units have 1. This is broad coverage, not proof that each large unit is covered deeply.

Recent-share references would put human physiology near 2.6 and genetics near 3.1 questions. Test 1 has 2 and 3. Structural organisation has 2 versus about 1.3; this is a modest one-paper variation. The bigger issue is that several Biology questions are easier than their labels.

Chemistry needs more variety in representation

Chemistry | Filled bars: Test 1. Open bars: recent current-scope share × 15.



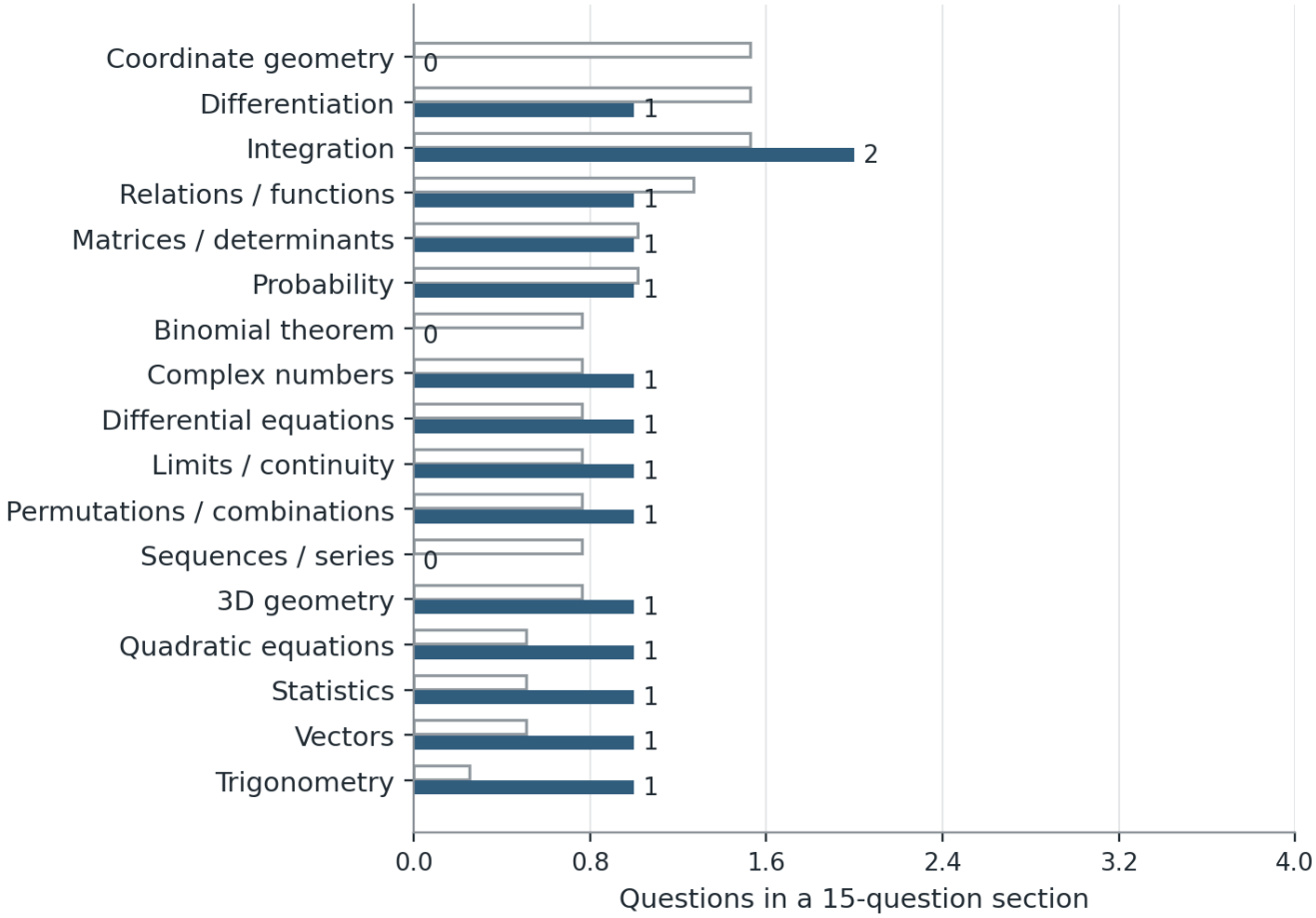
Denominator: 55 recent current-scope questions. Fractional references are not prescribed quotas. Each item counts once by its primary family.

Fourteen of 19 retained primary families appear. Carbonyl chemistry has 2 questions; every other selected family has 1. Atomic structure, d/f-block, hydrocarbons, halo compounds and alcohols/phenols/ethers have no primary slot.

Omitting five families in a 15-item section is not automatically a failure. However, organic principles are important in recent PYQs, and the adjacent Q24/Q25 ketone-to-alcohol chirality contexts narrow the range of organic thinking. Add structural interpretation by replacing or strengthening content at stable IDs.

Mathematics omits a recent leading family

Mathematics | Filled bars: Test 1. Open bars: recent current-scope share × 15.



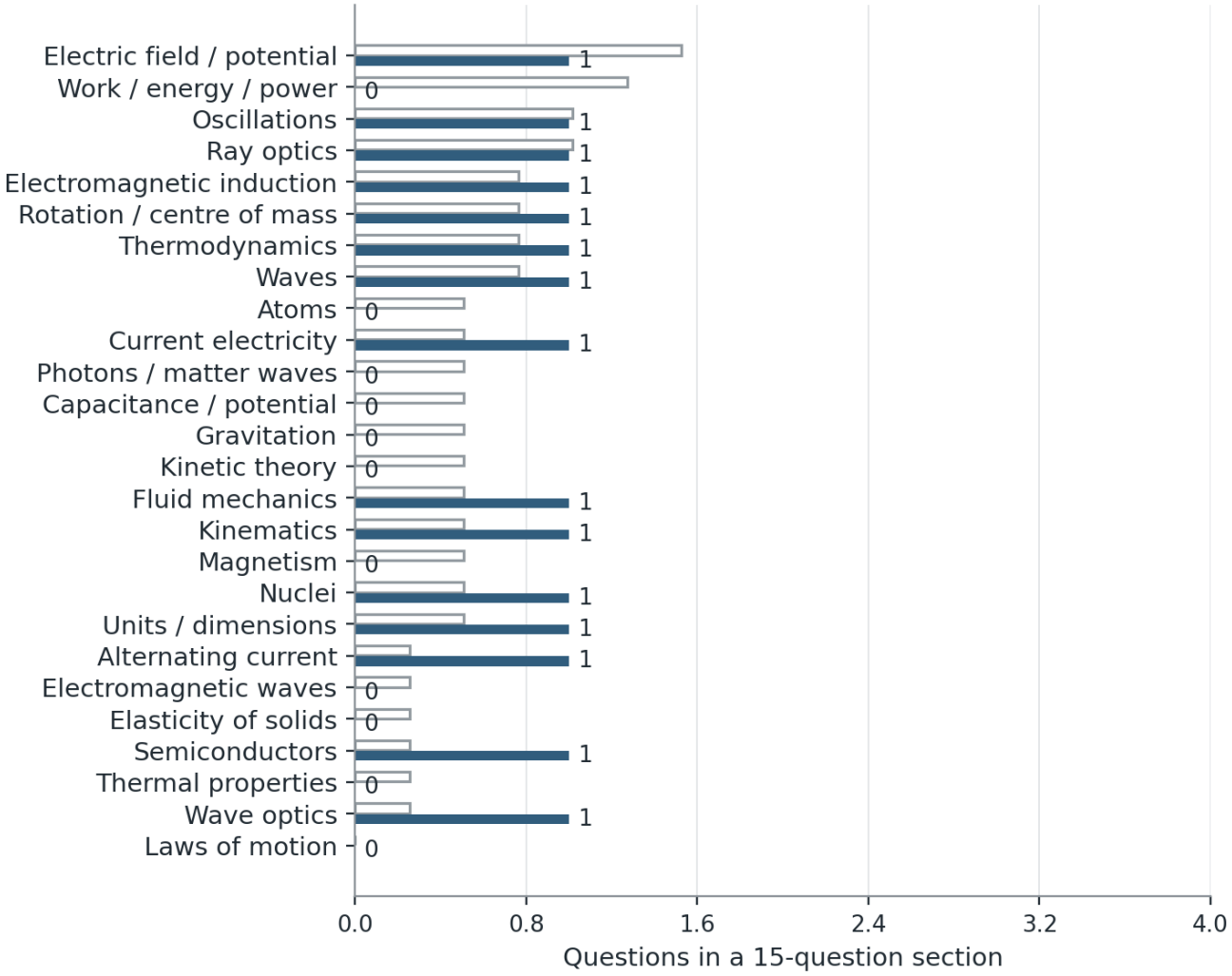
Denominator: 59 recent current-scope questions. Fractional references are not prescribed quotas. Each item counts once by its primary family.

Fourteen of 17 primary families appear. Integration has 2 questions. Coordinate geometry, binomial theorem and sequences/series have no primary slot. Vectors and 3D geometry do not replace the missing plane-coordinate/conics family.

Coordinate geometry has 6/59 recent current-scope items, tied with differentiation and integration. Replace an existing repeated or lower-priority slot with an original coordinate/conic problem at the same ID. The absence is a sample-balance concern, not a violation of an official chapter quota.

Physics is broad, with clear rotating omissions

Physics | Filled bars: Test 1. Open bars: recent current-scope share × 15.



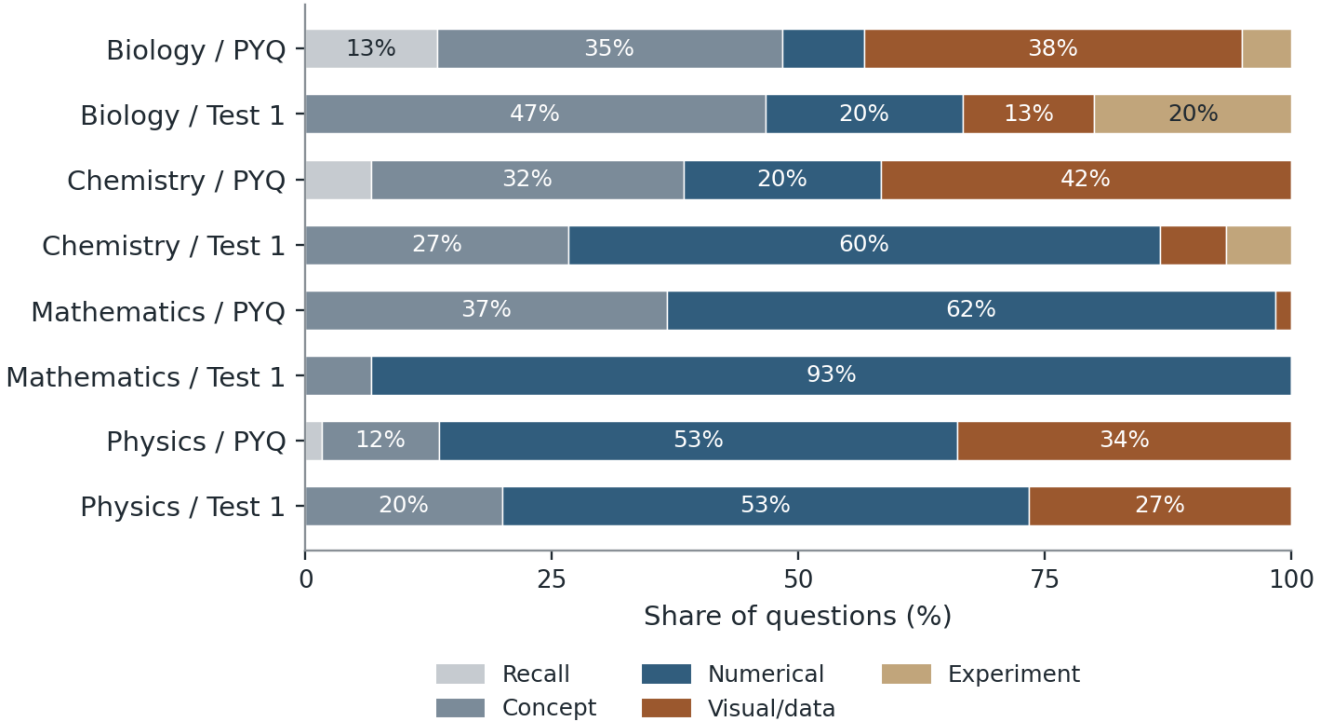
Denominator: 59 recent current-scope questions. Fractional references are not prescribed quotas. Each item counts once by its primary family.

Fifteen of 26 primary families appear, each once. Seven Class 11 and eight Class 12 primary topics are represented. A 15-item section cannot exhaust 26 families.

Work/energy, capacitance, magnetism, gravitation, thermal properties, kinetic theory, matter waves and atoms lack primary slots, among others. Energy reasoning is still used in Q49/Q53/Q59. Count these as secondary connections, not full primary-chapter coverage. Retain a series-wide rotation.

The style gap is larger than the chapter-count gap

Manual recoding uses the same task definitions as the recent PYQ ledger.



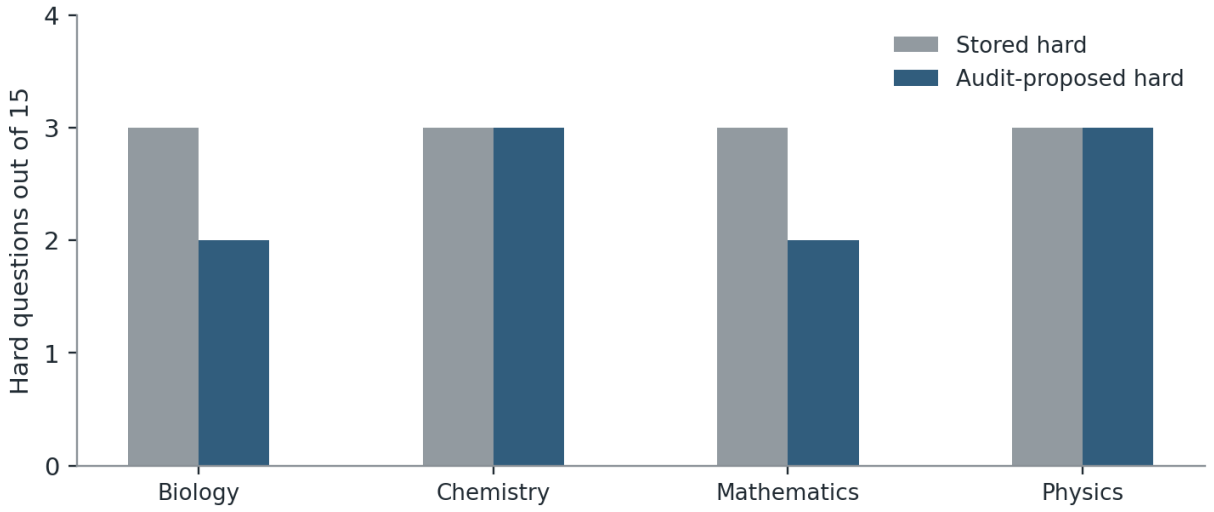
PYQ denominators: 60/60/60/59 in B/C/M/P, including scope-review items. Test 1: 15 each. Visual/data combines structures, diagrams, graphs and tables; it is not an image-file count.

Comparison	PYQ recent	Test 1	Audit implication
Chemistry numerical	12/60 (20.0%)	9/15 (60.0%)	Add structure/reagent interpretation; do not only add decorative PNGs
Maths numerical	37/60 (61.7%)	14/15 (93.3%)	Add conceptual domain/case/geometry reasoning
Biology visual/data	23/60 (38.3%)	2/15 (13.3%)	Use a real table, pedigree or labelled biological observation
Physics visual/data	20/59 (33.9%)	4/15 (26.7%)	Broadly closer; keep the meaningful graphs and circuit

The categories are editorial. Q24 is coded as structure reasoning, while its JSON format is conceptual. Printed chemical formulae alone are not automatically counted as visual tasks.

Difficulty labels currently overstate some items

Stored labels pass the project count rule; the content does not yet justify every label.



Set	Easy	Medium	Hard
Stored Test 1	9	39	12
Audit proposal	18	32	10
Recent PYQ tags, n=239	35	146	58

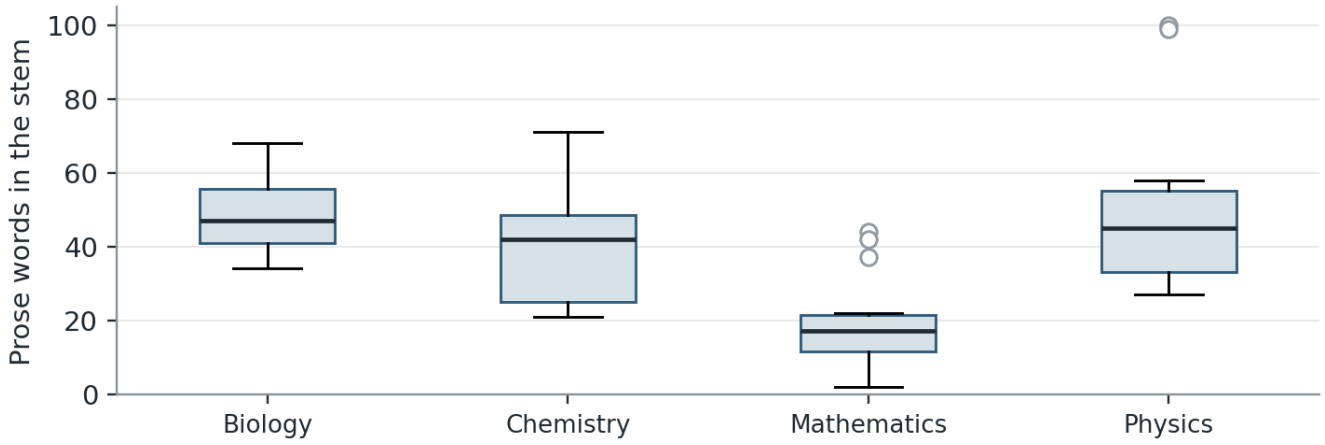
Medium to easy: Q5, Q8, Q12, Q14, Q21, Q27, Q29, Q51 and Q52. Their stems either directly supply the mechanism or leave a short familiar calculation.

Hard to medium: Q4 and Q33. Q4 can use the supplied ab frequency directly; Q33 is elimination followed by completing the square. Both are useful questions, but the hard labels are debatable.

Do not simply change labels and claim compliance. If the fixed 9/39/12 mix is required, strengthen the affected content or replace selected items, solve them again and recheck the mix. The proposed 18/32/10 split is a review judgement, not a measured outcome.

Readability, distractors and avoidable cues

Long wording can explain assumptions, but should not substitute for reasoning difficulty.



Prose-word counts replace each delimited formula by one placeholder and count alphabetic words; mathematical reading time is not measured. Each box summarises 15 stems: middle line = median, box = middle half, points = outlying counts.

Option-length cue: among 15 prose-heavy questions with no delimited mathematics in any option and at least one 8-word option, the correct choice is uniquely longest in **8**: Q3, Q8, Q9, Q10, Q12, Q14, Q16 and Q25. This is a screening signal, not proof that a student can infer answers from length.

Item	What weakens it	What to improve
Q12	The binding assay explicitly states the defect that the answer repeats	Ask students to distinguish competing evidence patterns
Q14	Photosynthetic frog skin and a four-chambered heart are implausible alternatives	Use credible errors about ventilation and diffusion
Q21 / Q29	The relevant reaction rules or quantitative response are supplied	Leave a real chemical identification step
Q27	1.90 g is unnecessary once pure final carbonate mass is given	Require mixture inference, or classify it as easy

Keep useful assumptions in Q4/Q11/Q20/Q49/Q60. Removing necessary controls merely to shorten stems would create ambiguity.

Are the six images random?

No: each is tied to the question data or physical setup. Their cognitive roles differ.

Q	Correct content checked	Purpose	Verdict
9	pH values, membrane and ATP-synthase sidedness	Orient a chemiosmotic setup	Relevant setup sketch
11	900/600/300 bp bands and non-quantitative intensity	Read the observed digest pattern	Genuine representation task
46	Exact v–t points, zero crossing, axes and units	Read signed versus absolute area	Genuine graph task
53	Negative field, break at a and constant segment	Read work from a field graph	Genuine graph task
54	Rail polarity, four resistors, junctions and meter	Understand connected topology	Genuine circuit task
57	Two depths, two indices, coin and viewing medium	Organise the layered setup	Relevant setup sketch

Two different questions about necessity

If the image and its alt text disappear, does the current stem lose data? Yes for all six: the pH values, observed bands, graph shape, resistor topology or layer values are not fully repeated in the main stem.

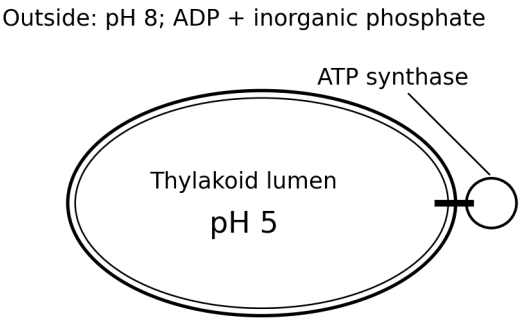
Does the diagram itself test a representation skill? Most clearly Q11/Q46/Q53/Q54. Q9/Q57 are helpful setup sketches whose data could be stated in one or two short sentences. They are relevant, but should not be counted as equivalent to a demanding graph-reading problem.

All six filenames match stable question IDs; all have alt text and register entries. No generated decorative illustration, answer-revealing result annotation or scientific mismatch was found. Detailed reproductions follow on pages 11–13.

Biology figures: correct, with a key reading limit

Reproductions from the unchanged pilot PNG files; the report adds no new question data.

Q9: Thylakoid gradient

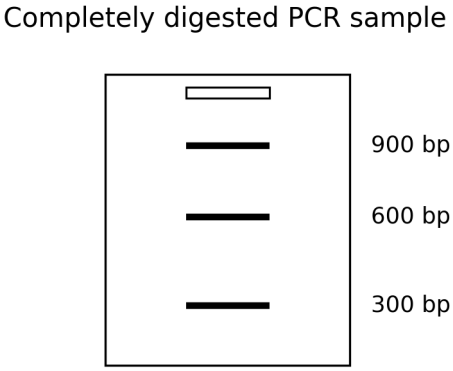


Darkness • sealed, proton-tight membrane

Accuracy: pH 5 in the lumen, pH 8 outside, and an outward catalytic head are consistent with transient ATP synthesis. There is no proton-flow arrow revealing the answer.

Purpose: Useful setup sketch. The two pH values and sidedness could be stated compactly in text; the sketch aids orientation rather than adding a new measured observation.

Q11: Restriction-digest gel



Band positions only; intensity is not to scale

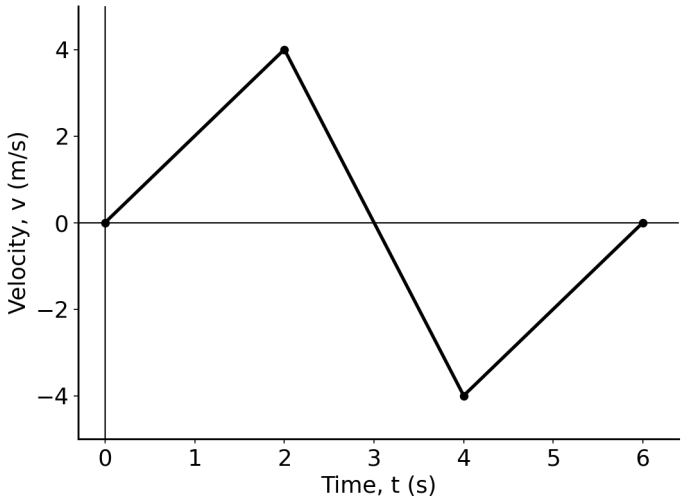
Accuracy: The 900/600/300 bp bands match complete digestion of the heterozygous product. Band positions and sizes are legible. The caption explicitly says intensity is not to scale.

Purpose: Genuine band-reading task. The equal-looking drawn bands must not be read as equal fluorescence. Keep the mass-proportionality assumption and non-quantitative-intensity caption.

Graphs: the shape is part of the problem

Axes, signs and key coordinates were checked against the stems and solutions.

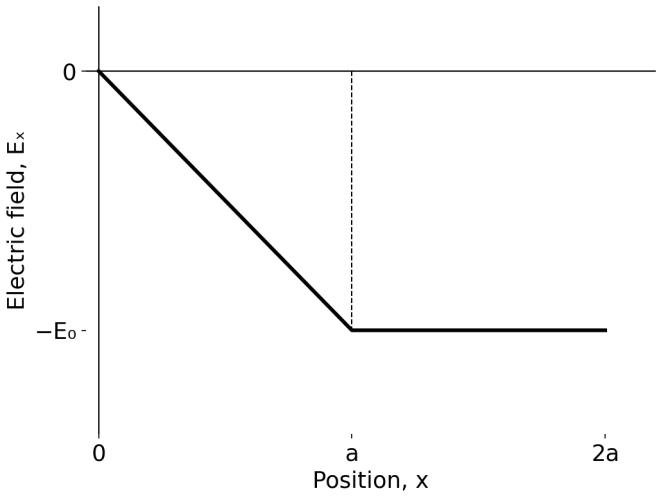
Q46: Velocity-time graph



Accuracy: Straight segments pass through (0,0), (2,4), (4,-4) and (6,0), so the central zero crossing is 3 s. Units and ticks are present; no area is shaded as an answer.

Purpose: Essential to the intended graph-reading skill. Signed areas cancel; absolute area gives 12 m, hence average speed 2 m/s and average velocity zero.

Q53: Electric-field graph



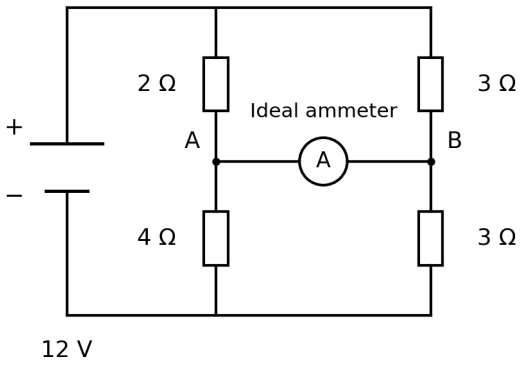
Accuracy: The field goes from zero to -E0 at a, then remains -E0 to 2a. The horizontal zero line and negative sign are correct. No stopping coordinate is marked.

Purpose: Essential to the intended area-reading skill. Work to a is -qE0a/2; the remaining 3qE0a/4 stops the charge at 7a/4, inside the drawn region.

Circuit and optics: keep the setup precise

The circuit carries topology; the layered diagram organises values that could also be stated in text.

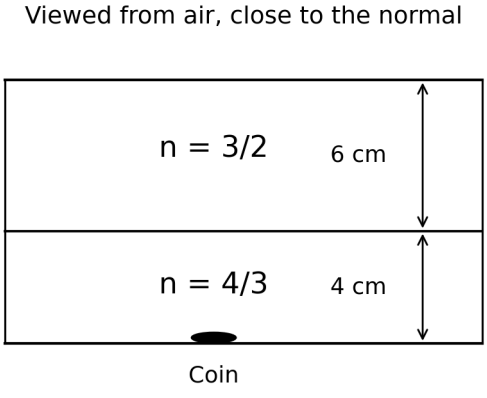
Q54: Connected resistor bridge



Accuracy: Supply polarity is top-positive; left resistors are 2/4 ohm, right resistors 3/3 ohm. Dots identify junctions A/B and the ideal meter connects them. The 12 V label no longer overlaps a resistor value.

Purpose: Essential topology diagram. The meter makes the midpoint potentials equal, but permits current. Connected-node KCL confirms 12/17 A from A to B, not zero.

Q57: Two parallel liquid layers



Accuracy: The upper layer is 6 cm at index 3/2; the lower layer is 4 cm at index 4/3; air is above and the coin is at the bottom. Dimension arrows delimit the correct layers.

Purpose: Useful setup sketch. The same data fit in two short text lines. No ray or virtual-image marker gives away the 7 cm apparent depth or its order invariance.

Keys and worked solutions: what the checks establish

No definite wrong key found in this review; checks are conditional on stated idealisations.

Item	Re-derived result	Main reasoning checked
B13	1/18	(2/3) family carrier × (1/3) unaffected population carrier × (1/4)
C20	0.74 V	Copper concentration 10^-12 M, then Nernst with n=2
C24	C8H14O2; 4 isomers	Reduction adds H2; acylation adds C2H2O; two centres, no terminal E/Z
M40	3/16	Choose two fixed points and derange the other four
M44	1/4	Split at sqrt(a); optimise at a=1/4 and compare endpoints
P49	4ω/3; work = $mR^2\omega^2/2$	Conserve angular momentum, not kinetic energy
P54	12/17 A, A to B	Connected-node KCL; midpoint potential 120/17 V
P60	3 micrometres	Smallest shared odd half-wave path is 1500 nm

Measured checks versus academic judgement

The validator finds 60 four-option items, key/explanation agreement, no duplicate normalised stems and no identical option texts within an item. Keys total 15 A, 15 B, 15 C and 15 D; no letter runs more than three times. The 45 computational probes pass, and all 328 delimited expressions render.

These do not prove semantic originality, optimal distractors, exact NCERT scope or real student timing. For example, Q26 should state initially equal hydrostatic pressures; Q56 should explicitly say the bar spans the rails perpendicular to them. Both stored results are consistent with the standard intended setup.

The 49-page review PDF matches its source/diagram checksums and contains 60 question headings and 60 solution headings. This audit does not reproduce every full solution; those remain in the pilot review PDF.

Biology: all 15 items reviewed

E/M/H below are the audit’s proposed tiers. Keys shown are the stored answers, not a revised answer key.

Q	Key	Tier	Finding / action
1	B	E	Keep. Generation and independence must be read together; introductory item.
2	D	M	Keep. Cambium and bundle position support the stated ordinary-growth prediction.
3	A	M	Keep. Substrate rescue is consistent with inhibition; wording avoids claiming proof.
4	C	M	Review tier: medium. The measured ab frequency can simply be squared; phase inference is not required.
5	B	E	Strengthen. Controlled conditions leave a familiar direct ADH response; likely easy.
6	A	M	Keep. Correctly separates plant respiration, assimilation and consumer respiration.
7	D	E	Keep. Explicit centromere convention removes chromosome/chromatid ambiguity.
8	C	E	Review tier: easy. The two fertilisation outcomes are direct; no endosperm calculation is needed.
9	A	M	Keep figure. Proton gradient and ATP-synthase orientation are correct; trim absolute distractors.
10	B	E	Keep. Active/passive distinction is sound; shorten the unusually long correct option.
11	C	H	Keep figure. Complete digestion and equal recovery justify band-mass inference; not equal band brightness.
12	D	E	Strengthen. Binding assay directly identifies the receptor defect, leaving little inference.
13	B	H	Keep. Both unaffected-person conditions are necessary; 1/18 follows.
14	A	E	Strengthen. Photosynthetic skin and a four-chambered frog heart are weak distractors; likely easy.
15	C	M	Keep. Opposite directions in replicate small populations support drift; a table would improve scanning.

Proposed tier count: 7 easy / 6 medium / 2 hard. 'Keep' means no specific correction was identified here; it does not replace human academic sign-off. Primary families are shown in the chapter comparison, pages 3–6.

Chemistry: all 15 items reviewed

E/M/H below are the audit’s proposed tiers. Keys shown are the stored answers, not a revised answer key.

Q	Key	Tier	Finding / action
16	D	E	Keep. Resonance is represented correctly; revise option-length cue.
17	A	M	Keep. Co charge balance and wavelength relation are consistent with the stipulated transition.
18	C	M	Keep. Correct unit conversion and temperature sign change; one short calculation chain.
19	B	M	Keep. Identical portions and independent assays remove mixture-scale ambiguity.
20	D	H	Keep. Ksp-controlled copper concentration and the two-electron Nernst factor are both needed.
21	C	E	Strengthen. The stem supplies the cold/warm rules, so the answer mainly copies the two amounts; likely easy.
22	A	E	Keep. Integer charge balance correctly interprets mixed valence; appropriate easy item.
23	B	M	Keep. Additional time is correctly distinguished from total time; supplied logarithms are adequate.
24	C	H	Keep with scope review. Formula tracking and two stereocentres are sound; explicitly map geometric/optical isomerism.
25	D	M	Replace one of Q24/Q25 at its stable ID: consecutive ketone/alcohol chirality contexts narrow organic coverage.
26	B	M	Clarify initial equal hydrostatic pressures. The osmotic result is correct under that standard assumption.
27	A	E	Review tier: easy. Final residue is already pure carbonate; the initial mixture mass is not needed.
28	D	M	Keep. Isoelectronic trend is unambiguous; use charge superscripts in all displayed options.
29	C	E	Strengthen. Both products and the two-silver stoichiometry are supplied; the remaining arithmetic is very short.
30	A	H	Keep. Strong-acid stoichiometry precedes the buffer ratio; approximation and final pH are sound.

Proposed tier count: 5 easy / 7 medium / 3 hard. 'Keep' means no specific correction was identified here; it does not replace human academic sign-off. Primary families are shown in the chapter comparison, pages 3–6.

Mathematics: all 15 items reviewed

E/M/H below are the audit’s proposed tiers. Keys shown are the stored answers, not a revised answer key.

Q	Key	Tier	Finding / action
31	C	E	Keep. Determinant and inverse trace are correct; appropriate short starter.
32	B	M	Keep. Ordered spacing and arrangements are counted once; enumeration confirms 96.
33	D	M	Review tier: medium. Two substitutions and completing the square give the endpoint minimum.
34	A	M	Keep. Intersecting loci and the signed product require interpretation rather than modulus alone.
35	C	M	Keep. Reflection symmetry plus substitution is valid; usable medium calculus item.
36	D	M	Keep. Plane reflection and origin distance are distinct; vector check agrees.
37	B	M	Keep. Both cusp locations must be cancelled; a=b=-2 is sufficient.
38	A	M	Keep. Population-variance convention is explicit; between-group variation is included.
39	D	E	Keep. Rationalisation is short and unambiguous; appropriate easy item.
40	C	H	Keep. Exactly two fixed points requires the four-key derangement count; exhaustive check agrees.
41	A	M	Keep. Principal-value quadrant is correctly enforced; option D needs proper inverse-tangent typography.
42	B	M	Keep. Separation, partial fractions and initial condition are valid; lower-medium demand.
43	C	M	Keep. Strict root location needs f(1)<0, not just the discriminant.
44	D	H	Keep. Split point moves with the parameter; optimisation and endpoints are correct.
45	B	M	Keep. Bilinear cross product gives the correct area; no picture is necessary.

Proposed tier count: 2 easy / 11 medium / 2 hard. 'Keep' means no specific correction was identified here; it does not replace human academic sign-off. Primary families are shown in the chapter comparison, pages 3–6.

Physics: all 15 items reviewed

E/M/H below are the audit’s proposed tiers. Keys shown are the stored answers, not a revised answer key.

Q	Key	Tier	Finding / action
46	A	M	Keep graph. Signed and absolute areas differ; zero crossing and scale are correct.
47	C	E	Keep. Dimensions of b follow term-by-term consistency; correct easy classification.
48	B	M	Keep. Radius-to-area factor is squared before Bernoulli is used.
49	D	H	Keep. Internal powered work is clearly distinguished from external axial torque; long stem merits timed review.
50	A	M	Keep. First-law balance and Carnot limit agree; no impossible engine is implied.
51	B	E	Review tier: easy. Maximum extension follows directly from equilibrium extension plus amplitude.
52	C	E	Review tier: easy. Two trial frequencies resolve the beat ambiguity quickly.
53	D	M	Keep graph. Negative field area decreases positive-charge kinetic energy; turning point lies within the plot.
54	B	H	Keep circuit. Meter shorts the midpoints; connected-node KCL confirms 12/17 A from A to B.
55	A	E	Keep. Alpha/beta-minus bookkeeping is correct and deliberately easy.
56	D	M	Clarify that the bar spans the rails perpendicular to them. The standard motional-emf result is correct.
57	C	M	Keep as useful setup sketch. Paraxial depth sum is correct and layer-order independent.
58	A	M	Keep. Reactive voltages oppose; 10 V rms and 16 W real power are correct.
59	B	M	Keep. Mean-square voltage, not mean voltage squared, gives the half-wave power ratio.
60	D	H	Keep. Initial pi reversal changes integer-wave conditions to odd half waves; 3 micrometres is consistent.

Proposed tier count: 4 easy / 8 medium / 3 hard. 'Keep' means no specific correction was identified here; it does not replace human academic sign-off. Primary families are shown in the chapter comparison, pages 3–6.

What must happen before release?

Keep the same IDs and image links. This report audits the draft; it does not silently revise it.

Priority	Concrete action	Completion evidence
1. Content calibration	Strengthen the nine medium-to-easy items; review hard labels Q4/Q33	Fresh solutions and a justified 9/39/12 split, if that policy is retained
2. Maths coverage	Use an existing slot, such as one repeated integration slot, for coordinate/conic reasoning	Coordinate family represented without expanding past 60
3. Style and distractors	Add real structure/table interpretation; address Q12/Q14/Q21/Q27/Q29 and Q24/Q25 overlap	Reasoning survives option elimination; no decorative image quota
4. Setup clarity	Add initial-pressure condition to Q26 and bar orientation to Q56	One clear intended model per question
5. Human trial	Obtain subject review plus a timed student attempt	Recorded correctness, ambiguity, timing and difficulty feedback
6. Target-year rules	Verify the 2027 syllabus and brochure before claiming compliance	Official source and date recorded

Sources and limitations

Official 2026 sources: IAT examination page and Information Brochure, sections 19/23, rechecked 29 October 2026. Project sources: canonical JSON, six PNGs, stored syllabus, 60-slot plan, PYQ ledger/metrics, validator, arithmetic and render reports.

PYQ limits: no recovered 2020 paper; 2025 English has 59 questions; 2026 is a provisional reordered key. Difficulty/style labels are editorial. No score/rank, causal trend, forecast accuracy or measured discrimination is established.

Audit calculations and all 60 item comments are saved in output/iat_reports/IAT_Test_1_Audit_Data.json. The audited source hash is on page 2. Final verdict: structurally sound pilot; revise for PYQ-style realism, then obtain human release approval.